Grant Description: NSF TTP-P: Video calling tools to enhance real-time alignment for remote teams

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Abstract
Teams that include both autistic and non-autistic members ("mixed-neurotype" teams) often experience challenges with communication, due to mismatches in communication styles and preferences. Repeated misunderstandings erode team trust and productivity over time and burden autistic team members with increased cognitive effort. In this NSF grant proposal, submitted to the Translation to Practice (TTP) program in January 2026, we envision tools for video calling that facilitate alignment in mixed-neurotype team meetings, ultimately improving employment outcomes for autistic people. We aim to investigate behavioral signals that reliably indicate conversational alignment or breakdown, and develop AI tools to detect these signals in real time during meetings and intervene to help resolve misunderstandings. Since NSF will not currently fund research on disability, we framed the proposal to address team communication more broadly, since all teams have at least some diversity in cognitive and communication styles. But we included critical grant activities to address the needs of autistic people and mixed-neurotype teams: a participatory design process with autistic and non-autistic stakeholders, and evaluation of the technology with real mixed-neurotype teams.

Vision
Teams that include both autistic and non-autistic members ("mixed-neurotype" teams) often experience challenges with communication (Crompton et al., 2020; Heasman & Gillespie, 2018; Milton, 2012; Wadge et al., 2019; Williams 2021). Mismatches in communication styles and preferences lead to misunderstandings, which erode team trust and productivity over time and directly burden autistic team members with increased cognitive effort (Lewis et al., 2024; Morris et al., 2015; Rebholz, 2012). Remote communication through video calls offers a lot of advantages for autistic workers, but it can also exacerbate some communication challenges (Das et al., 2021; Lewis et al., 2024; Tomczak et al., 2022; Zolyomi et al., 2019). With this grant proposal, we envision tools for video calling that can facilitate alignment in mixed-neurotype communication, improving employment outcomes for autistic people. We focus particularly on teams—a critical unit of collaboration and communication for a lot of knowledge work.

This proposal was submitted in January 2026 to the NSF Translation to Practice (TTP) program. (The proposal is still under review as of May 1, 2026.) Since April 2025, NSF has deprioritized research on "protected characteristics" including disability. Their current statement of priorities states that NSF-funded research "must aim to create opportunities for all Americans everywhere," and that "research projects with more narrow impact limited to subgroups of people based on protected class or characteristics do not effectuate NSF priorities." With that limitation in mind, we framed the proposal to address team communication more broadly, since all teams have at least some diversity in cognitive and communication styles. Under this framing, differences in neurotype are a quintessential example of the individual variation that exists in all teams; mixed-neurotype teams can be a "stress test" for universal tools. We included critical grant activities to ensure that the tools are helpful for autistic people and mixed-neurotype teams: a participatory design process with autistic and non-autistic stakeholders, and evaluation of the tool with real mixed-neurotype teams.

Research questions

We propose to develop Tools for Enhancing Alignment in Mixed-Neurotype Synchronous Communication (TEAM-Sync), a set of AI-powered features for video calling platforms to provide real-time feedback and support, surface misunderstandings as they occur, and reduce the cognitive burden and fatigue of video calls. This goal requires us to address research questions about human conversation behavior, human-computer interaction, and team dynamics:

1. What behavioral signals reliably indicate conversational alignment or breakdown? How do these differ across individuals and neurotypes? Can we use individual baselines to account for this variation when modeling conversation behavior?
2. What types of features and user interfaces for video calls help facilitate alignment during team meetings, particularly for mixed-neurotype teams?
3. Do tools to facilitate alignment impact teams by (a) changing observable conversation behaviors, and (b) improving perceptions of team trust, psychological safety, cohesion, and productivity?

Core activities

Grant activities build toward a scalable implementation of TEAM-Sync. Commercialization is an important component of the vision and grant: our tools will have a better chance to improve outcomes for neurodivergent people if they are broadly adopted by employers.

1. Develop novel approaches to analyze conversation behavior in real time, to track team alignment and detect signals of breakdowns in shared understanding.
2. Design video calling features to improve team alignment through an iterative, human-centered participatory design process with continuous feedback from autistic and non-autistic stakeholders.
3. Engineer a production-ready system (TEAM-Sync) to translate stakeholder-validated concepts from the participatory design process into a robust, scalable, and stable platform.
4. Evaluate the impact of TEAM-Sync with professional mixed-neurotype teams, assessing effects on team alignment, inclusion and participation of team members, cognitive effort and meeting fatigue, and team satisfaction and psychological safety.

5. Validate market fit and develop a commercialization strategy by testing and refining our hypotheses about the customer ecosystem and the value proposition of our technology.

Evaluation methods

We will evaluate the impact of TEAM-Sync on team communication in a study with 30 professional mixed-neurotype teams, recruited through partnerships with organizations that employ, recruit, or support neurodivergent people. Using a *switching replications* design, each team will record their normal work meetings at three time points over the course of 12 weeks, with the timing of access to TEAM-Sync staggered across two conditions. This design gives every team both a control period and an intervention period, strengthening internal validity.

We will analyze the meeting recordings to measure changes in conversational behaviors. After each meeting, members will report their perceived cognitive effort, fatigue, and enjoyment. In addition, team members will complete surveys before and after the study measuring team psychological safety, trust, satisfaction, and communication behaviors. We will assess how changes in conversation behavior relate to self-reported meeting experiences and team dynamics, and how the TEAM-Sync intervention affects autistic and non-autistic team members and the team as a whole.

Success metrics

To measure the success of our research and technology development, we will use the following quantitative and qualitative metrics:

- Model performance: Accuracy, precision, and recall of the multimodal machine learning models for detecting at least two validated conversation measures (Activity 1)
- System reliability: Crash-free session rates during the evaluation study (Activity 4)
- Stakeholder engagement: Successful recruitment of 25 co-design participants (Activity 2) and 30 mixed-neurotype teams for the evaluation study (Activity 4), including MOUs with organizations
- Improvement in team communication: Multiple quantitative measures of impact on team communication in the evaluation study (Activity 4)
- Feature adoption: Frequency of use of specific features by teams in the evaluation study (Activity 4)
- Stakeholder feedback: Feedback from design workshop participants (Activity 2), testers (Activity 3), Advisory Board members, and customer interviews (Activity 5) to determine whether TEAM-Sync meets expectations and addresses user needs.

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